Mapkeeper AI: Service Architecture and API Usage Plan

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1. Development Motivation, Problem Definition, and Purpose

Our team, "Mapkeeper," started with a field-oriented inquiry into how advanced AI technology can practically solve the problems faced by local alleyway commercial districts and small business owners. Understanding the actual difficulties experienced by micro-enterprise owners who protect our local neighborhoods—and grasping the essence of these problems—was the starting point for all our solutions. To confirm this, we stepped out of the lab and directly into the field.

1.1 Field Survey of Local Alleyway Businesses and Preliminary Research

During the previous Ajou University Paran Semester project, 'Dining Table Connection Group,' which aimed to revitalize local businesses, we conducted face-to-face field surveys by directly visiting about 150 restaurants in front of Ajou University. Through this process, we witnessed a severe disconnect between rapidly advancing technology and the reality of local businesses. We confirmed the serious problems faced by the owners with objective data as follows.

1.1.1 Severe Digital Marginalization in Local Businesses (Arosaegil Project)

We visited 150 restaurants in person to encourage participation in a Naver Place keyword registration project ('Arosaegil Matjib'). As a result of our observation, most of the restaurant owners (mainly in their 50s and 60s) experienced extreme difficulty simply operating a smartphone. Participation rates via text notices or online guides were 0%, failing to induce any engagement.

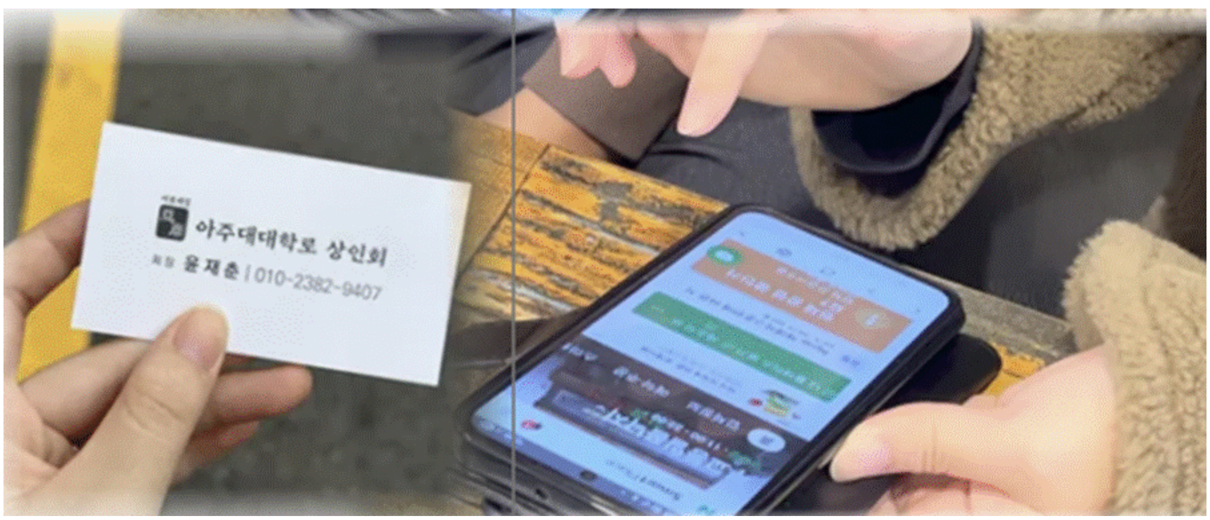
However, when we applied a 'face-to-face agency' approach, where a student directly took the owner's smartphone, installed the app, and set it up for them, 41 restaurants finally participated. Through this, we painfully realized that for elderly small business owners, screen-based 'operation guides' are meaningless, and an intuitive interface (VUI, automation) where someone (or something) handles it for them is absolutely essential.

1.1.2 Fatal Decline in Brand Awareness Due to Lack of Online Information

Investigating 'Mr. Chef,' a restaurant boasting excellent taste with a 20-year tradition, revealed a shocking phenomenon: 42.7% of current students responded that they had "never been there" due to a lack of online information management. However, after creating menus targeted at specific demographics (e.g., international students) and conducting storytelling marketing, awareness increased by 21%, proving that online visibility is directly tied to the survival of offline restaurants.

1.1.3 Explosive Sales Growth from Data Organization and SEO Optimization

We confirmed the effects of systematic management through numbers. By solving the duplicate registration problem on Naver Place and organizing missing essential information (business hours, menu photos), 'SsanQ' saw a 13.9% increase in September sales and an 18.2% increase in October sales compared to the same period last year, resulting in a total sales increase of 16.05%. For 'Hokago,' by adjusting menu unit prices and concentrating on Search Engine Optimization (SEO) marketing through Reels and blog exposure, the cost-of-goods-sold ratio improved by 4%p, and the daily average net profit soared by an incredible 29.2% compared to the previous month. In other words, we confirmed that even micro-enterprises can achieve immediate and explosive sales growth simply by receiving systematic online map management.



<(Figure 1) Field Survey Photos>

Through field surveys and research, we discovered a dilemma: while online map platform management is a lifeline directly linked to small business owners' sales (16~29% increase), the digital barrier to entry is simply too high for the owners to perform this themselves, as confirmed by our 150-store visit. Based on this vivid field data, we began planning 'Mapkeeper AI,' an artificial intelligence project that will act as an agent to perform the 'information integration and modification' and 'SEO marketing' that students previously did manually.

1.2 In-depth Analysis of 'Information Asymmetry' and Digital Marginalization in Local Businesses

As offline word-of-mouth culture in university areas was cut off after COVID-19, the commercial exploration route for those in their 20s and 30s transitioned 100% online (map apps, review platforms). However, our face-to-face survey showed that the digital marginalization of micro-enterprises was acting as the root cause of local business stagnation. The specific pain points can be classified into four dimensions:

1.2.1 High UI/UX Barrier to Acquiring Administrator Privileges

Most restaurant pages are left unattended, registered arbitrarily by customers rather than the owners. For an owner to edit information, they must claim 'administrator privileges.' However, complex procedures such as photographing and uploading a business registration certificate, dealing with customer service chatbots, and multi-step identity verification cause elderly owners to give up without even starting.

1.2.2 Management Abandonment and Information Inconsistency Due to Multi-Platform Fragmentation

Consumers use various maps like Naver Place, Kakao Map, and Google Maps, but owners must manually edit information in each different app and admin page. Information inconsistency deepens; for example, prices are raised due to increased food material costs, but old prices remain online, causing arguments with customers, or rest days are not updated, leading to bad reviews from students who made a wasted trip.

1.2.3 Lack of Review Communication and Total Absence of SEO Knowledge

Owners have no idea what keywords to use (e.g., 'Ajou University cost-effective solo dining') to rank high in search results, nor do they know how to respond to negative reviews. As a result, despite having sufficient taste and service competitiveness, they are pushed down to the very bottom of online searches by large franchise restaurants.

1.2.4 Loss of Information and Consumer Attrition (Resulting Aspect)

Consumers in their 20s and 30s, including the "untact" (non-face-to-face) generation of students, aim for 'failure-free consumption' within limited budgets. Local alleyway restaurants with blurry menu photos, uncertain business hours, and no recent reviews are stigmatized as "unmanaged = bad tasting" and are completely excluded from potential visit lists, creating a severe information asymmetry problem.

1.3 Root Cause Analysis Based on Design Thinking

Moving beyond the one-dimensional assumption that "they can't use smartphones because they are old," we redefined the structural causes of the digital divide among small business owners found in the field using design thinking methodologies.

5 Whys Technique: Tracing from Surface Phenomenon to Structural Cause

Phenomenon: Local restaurant owners give up on and abandon online store management.

- **Q. Why 1:** Why do owners give up on online store management?
  - **A.** Because the authorization application process and information modification UI are overwhelmingly complex.
- **Q. Why 2:** Why does the information modification process feel so complex?
  - **A.** Because platforms demand document (business registration) uploads or multi-step text authentication tailored to PC environments in the name of 'security' and 'standardization.'
- **Q. Why 3:** Why are document uploads and multi-step authentication such massive barriers for owners?
  - **A.** Because owners only use smartphones for simple communication (KakaoTalk) or video consumption (YouTube) in their daily lives, and they have absolutely zero learning experience with text-heavy administrative UI/UX that requires navigating complex depths.
- **Q. Why 4:** Then why can't they learn new digital device operations or UIs?
  - **A.** Because amidst the exhausting physical labor environment of moving between a hot, greasy kitchen and the dining hall for 12 hours a day, they absolutely lack the physical time and psychological capacity to sit down and study an unfamiliar system.
- **Q. Why 5:** Why is there such an extreme disconnect between the owners' labor environment and the systems?
  - **A.** Because current store management platforms are designed based on 'white-collar administrators who can sit in front of a PC and focus intently on a screen,' completely excluding the work environment and cognitive characteristics of 'field workers (small business owners) who work on their feet all day and are accustomed to intuitive verbal communication.'

AEIOU Observation: Analysis of Small Business Owners' Physical Work Environment

By analyzing the actual working environments of owners observed during visits to 150 restaurants, we clarified the limitations of the existing visual/touch-centric UI.

- **A (Activities):** Owners perform constant, physical multi-tasking: taking orders, cooking, serving, clearing tables. The only time they can look at a smartphone is during break times or after work when fatigue is at its peak.
- **E (Environments):** The kitchen deals with fire, water, and oil. With wet hands or while wearing sanitary gloves, it is physically very uncomfortable to finely touch small text screens on a smartphone.
- **I (Interactions):** Within the restaurant, communication between the owner and customers, and the owner and staff, is 99% 'verbal (voice).' Intuitive and immediate verbal instructions (e.g., "Auntie, one soup here," "Prepare green onion pancakes since it's going to rain tomorrow") are the most familiar mode of interaction.
- **O (Objects):** The main objects they handle are food ingredients, cooking utensils, and POS

machines—physical tools that show immediate reactions. This is far from complex software structures where one must dig through hidden menus.

- **U (Users):** They are an aging population (50s-60s+) who struggle to read small print due to presbyopia. When learning something new, they prefer a conversational information acquisition method of asking a person directly and listening to the answer, rather than reading printed text (manuals).

As a result of the analysis, the digital marginalization of small business owners is not simply a lack of capability but stems from an extreme mismatch between platform design and the field work environment. Therefore, this problem can never be solved with existing app (UI) methods that rely on sight and touch; an 'audio (VUI)'-based agent tool, which utilizes the most familiar interaction method for owners, is inevitably required.

1.4 Limitations of Existing Solutions: The Structural Contradiction of 'One-Dimensional Education'

Currently, public institutions (Small Enterprise and Market Service) and private platforms (Naver Business School, Kakao, etc.) are spending massive budgets on large-scale 'digital capacity-building education,' but this has four fatal limitations in the reality of local commerce.

1.4.1 The Absolute Limit of Education Coverage (The Paradox of Scale)

As of 2024, there are 2.52 million representatives in their 50s (41.2%), but the scale of educational support for a year amounts to only tens of thousands. The idea of educating a long-tail market of millions with a small number of instructors or offline infrastructure is physically impossible.

1.4.2 The Absence of Physical Time

Even if owners—who play multiple roles (cooking, serving, ordering supplies, tax accounting, etc.) for 12 hours a day—learn how to use the platforms through education, they absolutely lack the physical and mental time to consistently apply it to marketing and store management.

1.4.3 Short-Term Events and the Forgetting Curve

Store information updates happen intermittently. Once field mentoring ends, owners get caught up in their livelihood, quickly forget the complex operations, and revert to a state of management abandonment.

1.4.4 The Cycle of Platform UI Updates and Re-learning

Even if they manage to learn a specific operation, a single update to the platform's user interface (UI) or security authentication invalidates their learning, trapping them in a vicious cycle of having to learn new device operations all over again.

1.5 Target Personas

Based on field research and in-depth analysis, we derived four personas as the direct stakeholders who will use this service.

[Main Persona 1: Digitally Marginalized Small Business Owner]

"Since my son went to the military, I don't even dare to fix our store information on my smartphone. It only gets touched when he comes out on leave, but he's busy job hunting too, so our store's online management is completely abandoned."

- **Name / Age:** Kim Young-hee (61, Female, running a restaurant 4 minutes from Ajou Univ. main gate for 20 years)
- **Situation & Pain Points:** Uses her smartphone only for KakaoTalk and YouTube. Relied on her college-aged son for Naver Place, but it has been 100% neglected since his enlistment. Raised soup prices due to ingredient costs but doesn't know how to reflect it online, leading to frequent arguments with students at checkout. Sales have dropped about 28% since COVID-19 due to losing ground to franchise restaurants. Has zero marketing capability.
- **Top Need:** Needs a function where, without relying on busy children or learning complex operations, she can simply say, "Change it to open until 9 tomorrow," or "Write a nice reply to the review," and all three platforms (Naver, Kakao, Google) are managed simultaneously, with AI automatically pointing out SEO ranking factors.

[Main Persona 2: Hands-on Owner Blocked by Platform Admin Barriers]

"I know how to use a smartphone. But why is it so hard to just find the Naver admin rights? The store is crazy busy, and after wrestling with the Naver Place chatbot that repeats the same robotic answers, I eventually gave up. Meanwhile, reviews are scattered everywhere, and customers make wasted trips looking at old info."

- **Name / Age:** Choi Jin-soo (39, Male, running a 'Youth Bubble Tea' shop for 15 years)
- **Situation & Pain Points:** Proficient with smartphones and delivery apps. However, he tried multiple times to claim admin rights to fix an error where his store was duplicated twice on Naver. Exhausted by mechanical macro replies ("Cannot find authorization") and complex administrative procedures, he gave up on updating the info.
- **Top Need:** When administrative errors occur (like privilege changes or duplicates), he wants a 'fixer' agent that goes beyond frustrating chatbots to immediately diagnose the problem and connect him to the exact solution or deep link all at once.

[Sub Persona 1: Information-Centric MZ Generation College Student]

"I don't have seniors to ask, and I can't take a risk going to a store that doesn't even have photos on the internet."

- **Name / Age:** Park Ji-hoon (23, Male, 3rd/4th-year Business major at Ajou Univ.)
- **Situation & Pain Points:** Due to non-face-to-face classes during early COVID-19, he lacks interaction with seniors and hears zero offline word-of-mouth information. With limited

allowance, he relies on map app searches for over 95% of his 5-6 weekly meals out. He unconditionally excludes old alleyway shops with no reviews or poor photos, assuming they are "out of business or unmanaged."
- **Top Need:** Wants reliable, up-to-date online information (business hours, accurate prices, clear photos, transparent review communication) even for small alleyway restaurants.

[Sub Persona 2: Job Seeker Maximizing Time Efficiency]

"It's truly deflating when I take a 20-minute round trip from the library to eat, only to find a piece of paper stuck to the door saying, 'Closed due to personal reasons.' My precious time..."

- **Name / Age:** Lee Soo-ah (23, Female, 4th-year at Ajou Univ.)
- **Situation & Pain Points:** Lives in the library preparing for jobs and certifications, splitting her time into minutes. Frequently experiences wasted meal times and energy because a store shown as 'Open' on the map app is closed for unannounced holidays or break times.
- **Top Need:** Wants the owner's real-time store situation (temporary closures, break times, sold-out ingredients) to be immediately reflected on map apps to fundamentally prevent wasted trips.

1.6 Development Purpose and Expected Effects

The ultimate purpose of this project is to fundamentally resolve information asymmetry in local commercial districts by developing an AI agent, 'Mapkeeper AI,' that can comprehensively control complex multi-map platforms (Naver, Kakao, Google) through a single voice command (VUI).

1.6.1 Small Business Owner Aspect (Breaking Barriers and Increasing Sales)

Drastically lowers the fear and fatigue of operating smartphones through Voice User Interface (VUI). Saves energy wasted on updating multiple platforms, and drives actual sales increases by turning the footsteps of customers in their 20s/30s away from franchises back to local shops using AI-based custom SEO strategies.

1.6.2 Consumer Aspect (Failure-Free Consumption)

Prevents wasted trips by providing accurate, up-to-date information, and vastly expands the right to explore and choose hidden alleyway restaurants that consumers previously avoided due to a lack of reviews or information.

1.6.3 Local Community Aspect (Ecosystem Revitalization)

Restores the tilted playing field of online promotion between large restaurants with massive marketing capital/dedicated staff and micro-restaurants using AI technology, building a healthy, coexisting local ecosystem.

1.6.4 Technical Aspect (AI Demonstration Model)

Presents a scalable reference for advanced artificial intelligence technologies (LLM, STT, Vision AI) as excellent 'appropriate technology' that completely changes the practical survival metrics of marginalized groups, going beyond simply enhancing advanced IT environments.

2. Development System Overview

2.1 Service Introduction and Core Values

- **2.1.1 Project Name:** "Mapkeeper AI" (맵지기 AI)
- **2.1.2 Reason for Naming:** It means an "AI keeper protecting the map." It is a warm name designed to present complex high-tech features—multi-channel integration, voice command control, and AI branding automation—in a simple, friendly way so that elderly owners can approach it without resistance.
- **2.1.3 Main Slogan:** "Manage your store all at once, with just a single word."
- **2.1.4 Sub Slogans:**
  - "Complex Naver, Kakao, and Google—AI manages them for you."
  - "University eats without digital exclusion, Mapkeeper AI protects them."

2.2 Core Functions and Solution Methods

Mapkeeper AI is not a passive, simple integration tool, but an intelligent proxy agent that recognizes the owner's voice and autonomously performs tasks.

2.2.1 Integrated Store Information Management Engine

Manages store information scattered across multiple platforms (Naver, Kakao, Google) on a single screen.

- **Integrated Data Structure Design:** Consolidates store information (hours, menu, prices) into a single standard structure.
- **Platform-Specific Processing:** The system separates and transforms data to match the format required by each platform, applying it simultaneously.
- **Update Status Check:** Easily displays via a visual dashboard which information is delayed or applied on which platform.

2.2.2 Voice Input-Based Command Processing (VUI)

Performs tasks solely through the owner's spoken words, without needing to navigate complex menu depths.

- **Speech-to-Text (STT):** Instantly converts the owner's colloquial speech into text.
- **Intent Classification & Entity Extraction:** Classifies the task to be performed (hours, photo registration, etc.) from the converted sentence and extracts key parameters (date, time).
- **Task Command Generation:** Automatically converts it into a structured command format that the internal system can process and forwards it to the backend.

2.2.3 Intelligent Review Analysis and Keyword Recommendation

- **Review Collection & Arrangement:** Consolidates customer reviews from multiple platforms.
- **Sentiment Analysis:** Analyzes the positive/negative trends of reviews to grasp the overall reaction.
- **Core Keyword Extraction:** Extracts frequently mentioned strengths or core promotional keywords (cost-effective, atmosphere, friendly, etc.).
- **Copy Recommendation:** Automatically generates and recommends store introductions or marketing copy based on the extracted keywords.

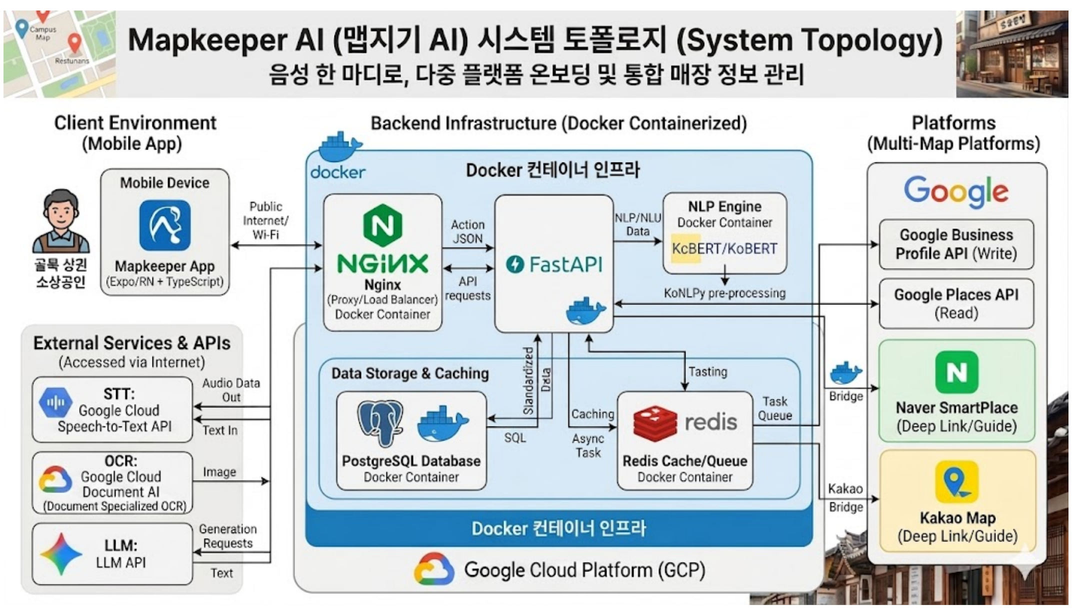
2.2.4 Authorization Acquisition and Initial Setup Support

- **OCR-Based Information Extraction:** By simply uploading a photo of the business registration without filling out complex forms, OCR automatically extracts the business name, registration number, etc., to help with initial store setup.
- **Procedure Guide & Help Chatbot:** The chatbot guides step-by-step through the complex administrative procedures that occur when requesting admin rights on each platform and provides deep links.

2.3 Detailed Implementation Plan and Architecture

2.3.1 Tech Stack and Multi-Platform Integration Architecture (API Strategy)

This project designed its architecture comprehensively considering market share and technical feasibility. The diagram is as follows.



<(Figure 2) System Architecture>

1) System Topology and Backend Architecture This system is designed with a Docker container-based microservices architecture to ensure scalability and the stability of asynchronous tasks.

- **Client (Mobile App):** Implemented with Expo (React Native) and TypeScript to provide an intuitive voice interface to the owner.
- **Backend Infrastructure (Docker Containerized):** Uses Nginx as a reverse proxy in a **Google Cloud Platform (GCP)** environment to distribute traffic, and FastAPI handles the main business logic.
- **Data Storage and Caching:** Fragmented store information by platform is standardized and stored in PostgreSQL. To prevent bottlenecks during multi-API communication, a Task Queue (asynchronous task queue) is operated using Redis.
- **External Service Integration:** Communicates with **Google Cloud Speech-to-Text API**, **Google Cloud Document AI (Document Specialized OCR)**, and **LLM API** modules. For information updates, it integrates Google Business Profile API (Write) and Places API (Read) dually.

2) Tech Stack Summary

- **Frontend:** Expo (React Native), TypeScript
- **Backend & DB:** FastAPI (Python), PostgreSQL, Redis
- **AI/ML:** Google Cloud Speech-to-Text API, Google Cloud Document AI, Hugging Face Transformers (KcBERT/KoBERT), LLM API
- **External API Integration:** Google Business Profile API, Google Places API
- **Deployment/Ops:** Docker, Nginx, GCP

3) Multi-Platform Integration Architecture (Google-Centric MVP Strategy) Domestic platforms (Naver SmartPlace) have high market share but present extreme administrative constraints for automated API control, such as requiring direct owner login, a limit of 5 edits per week, and a maximum 5-business-day review period. Therefore, the 1st MVP is implemented centering on the Google ecosystem, where real-time info modification and demonstration are perfectly possible via official APIs.

- **Write Path - Google Business Profile API:** Automates practical information updates of the owner's store (hours, menu edits, review responses) in real-time.
- **Read Path - Google Places API:** Collects detailed information and review ratings of the owner's store and competing stores in the surrounding commercial area.
- **Domestic Platform (Naver/Kakao) Defensive Strategy:** For items on domestic

platforms where direct API updates are fundamentally impossible, a separate 'Deep-link based guided update structure' is built to prevent owner drop-off, allowing them to jump directly to the actual modification page of each platform.

2.3.2 Intelligent Review Analysis Pipeline (NLP Tech Stack)

To go beyond simple frequency counting of reviews and generate precise commercial area analysis and recommendation copy based on context, a dual NLP pipeline is introduced.

- **1) Preprocessing & Keyword Extraction (KoNLPy):** Uses a morphological analyzer to handle unclear word boundaries and postpositions unique to the Korean language. Through this, core promotional noun tokens like "cost-effective," "hygiene," and "solo dining" are accurately extracted from reviews.
- **2) Sentiment Analysis & Semantic Grasp (Hugging Face - KcBERT/KoBERT):** Since morphological analysis alone cannot determine the positive/negative context of the whole sentence, a pre-trained language model is used. Considering the characteristics of restaurant reviews, which are short and heavily colloquial, KcBERT, trained on Korean comment data and boasting high domain suitability, is used as the primary sentiment classification model.
- **3) 3-Stage Data Learning Pipeline Design:** To secure initial reliability, competing store reviews in the surrounding area are collected via Google Places API, and fine-tuning is conducted with a large-scale Korean review sentiment dataset published on AI Hub. Finally, actual restaurant reviews near Ajou University are directly labeled to advance the model into a perfectly specialized model for the university commercial district domain.

2.3.3 Voice Command Processing Pipeline

To minimize the burden of digital input on elderly owners, it is designed as an 'Imperative VUI' that recognizes short commands clearly and connects them to actions, rather than requiring lengthy conversational responses.

- **Speech-to-Text (NAVER CLOVA Speech):** Adopts the Naver Clova STT engine, which shows top-tier transcription accuracy for recognizing short, colloquial Korean commands like "Change business hours to tomorrow until 9" or "Set green onion pancake photo as the representative menu."
- **Text Preprocessing & Entity Candidate Extraction (KoNLPy):** Cleans the sentence structure with a morphological analyzer to remove incomplete spacing or ambiguities included in the speech result, extracting noun/verb candidates.
- **Voice Command Intent Classification Model (KcBERT/KoBERT):** Accurately classifies which task intent category the user's speech falls into, such as 'modify business hours', 'change menu', 'register photo', or 'check reviews'.
- **Core Information & Slot Extraction:** Combines KoNLPy token analysis and rule-based logic

to extract core parameters (date, time, menu name, price, etc.) within the utterance. Finally, it automatically converts it into a structured JSON command object that the internal store management engine can process immediately.

2.3.4 Mapkeeper AI Core User Flow (4-Step Process)

The process from the user's utterance to actual reflection on multiple map platforms goes through the following 4 automated steps.

- **1) Voice Command:** The owner opens the app and speaks shortly and clearly in everyday language, like "Say we close at 8 PM just for today."
- **2) AI Analysis & Structuring (NLU/STT):** NAVER CLOVA STT transcribes the voice to text, and the NLU module (KcBERT) extracts the intent (UPDATE_OPERATING_HOURS) and slot (today 20:00). The system automatically converts this into an executable JSON format like: {"intent": "update_business_hours", "slots": {"date": "today", "end_time": "20:00"}}
- **3) Execution & Synchronization:** The FastAPI server assigns the received command to the Redis Task Queue, and calls the respective platform APIs (Google, Naver, Kakao) to asynchronously update the data in the background.
- **4) Completion & Feedback:** Once processed, 'Today's closing time: 20:00' is updated on the store dashboard, providing the owner with a visual completion notification.

<(Figure 3) Mapkeeper AI Core User Flow>

3. Related Technology Trends and Similar System Analysis

3.1 Analysis of Similar Systems and Products

Existing store management and commercial area solutions are skewed towards providing 'macro information' or are designed around 'complex features,' showing limitations in driving actual execution in local alleyway ecosystems. The pros and cons of major similar systems are analyzed below.

Comparison Item	Small Business 365 (State-led Big Data Platform)	Global Sync Tools (Yext, Uberall, etc.)	Mapkeeper AI
Similarity (Criteria)	Providing commercial data for small business owners (Similarity of purpose)	Integrating store info across maps (Similarity of technology)	**Custom Local Solution for Small Businesses**
Core Value	Macro data diagnosis of commercial areas	Global API synchronization of worldwide store info	**Real-time execution & management of local alley info**
Main Advantage	Credibility based on public/private data	Management efficiency for large franchise stores	**Overwhelming usability for owners in their 50s/60s**
Tech Features	Statistical analysis, BI dashboard reports	Large-scale bulk updates based on B2B APIs	**Voice Agent (VUI) & AI SEO Optimization Support**
Critical Weakness	(Cognitive Gap) Statistics cause cognitive overload for busy owners. No practical answer to "How do I fix my store's keywords today?"	(Failure in Korean Localization) Cannot break through the specifics of the Korean market (Naver/Kakao) which demands strict admin auth like business reg certs.	Dependency on Google API for 1st MVP. Requires optimization of bypass guide features like deep links.
<(Table 1) Similar Service Analysis>			

3.2 Research and Latest Tech Trends Related to the Development Topic

This project integrated the latest technology trends such as Multi-API integration, Natural Language Processing (NLP), and Voice User Interface (VUI) into its architecture to overcome fragmented platform environments and accessibility limitations for small business owners.

- **Multi-Platform Integration and Asynchronous Microservices Architecture**
  - Standardizing store info from various platforms into PostgreSQL and operating a Task Queue (Redis) to prevent bottlenecks during sync is a major trend in modern backend design.
  - To bypass administrative constraints of domestic platforms (Naver), an integration strategy combining real-time sync via official APIs (Google) and a domestic 'Deep-link based guided update structure' is being actively researched and developed.
- **Intelligent Review Analysis NLP Pipeline**
  - Technology using morphological analyzers (KoNLPy) to preprocess and extract core noun tokens in reviews for precise, context-based analysis beyond simple counting is actively used.
  - To overcome the domain specifics of short, colloquial restaurant reviews, research heavily utilizes deep learning models based on Transformers like KcBERT/KoBERT (pre-trained on Korean comments) to analyze sentiment and recommend marketing copy.
- **Imperative VUI Technology**
  - For elderly users burdened by digital operation, 'Imperative VUI,' which connects short, clear voice commands to actions instead of lengthy conversations, is emerging as an alternative tech.
  - The core trend is an AI pipeline merging high-performance STT engines (NAVER CLOVA) optimized for Korean colloquial commands with NLU models to classify intent and automatically convert core slots (date, time) into structured JSON command objects.

4. Originality and Differentiation

Mapkeeper AI is not a competing product fighting giant IT platforms, but an 'Intelligent Bridge' connecting advanced tech and 5060 small business owners, possessing three overwhelming unique traits.

4.1 'Omnichannel Bridge' Not Subordinate to a Specific Ecosystem

Large platforms (Naver, Kakao, Google) inherently cannot provide an integration feature that binds and syncs info from rival platforms due to market share competition. Mapkeeper AI secures a stable, independent, and neutral agent status that connects fragmented apps strictly from the 'owner's perspective,' without being tied to any specific IT ecosystem.

4.2 Intelligent Pipeline Embracing the 'Long-Tail Market'

Existing info registration and marketing management required 1:1 human consultants. Due to massive labor costs, millions of micro local businesses have always been neglected in an 'unprofitable blind spot.' Mapkeeper AI perfectly replaces this expensive 'close human consulting' with AI automation pipelines like STT engines and NLP classification models. Consequently, it achieves incredible innovation by expanding agency-level management to the entire neglected long-tail market at high efficiency and low cost.

4.3 'Target-Customized Zero-UI' Free from the Shackles of General UI

Existing store management apps (CatchTable, Toss, etc.) still require owners to type text within closed ecosystems and adopt complex visual/touch-centric UIs designed for younger generations and office workers. However, Mapkeeper AI deeply reflects the cognitive models of 5060 owners and the physical work environment of a kitchen, boldly eliminating the process of finding complex buttons on a screen. By bringing everyday, intuitive voice commands to the forefront as the core interface (Imperative VUI), it achieves an overwhelming accessibility and intuitiveness ('Zero-UI') where 'there is no need to even learn how to use the system.'

5. Development Method

5.1 System Components

Mapkeeper AI is an AI-based integrated management system supporting 5060 small business owners to easily manage store information using voice and photos. The system is designed with a 3-tier architecture to clearly divide roles and secure maintainability/scalability.

1. **UI/Controller Layer**
2. **Core Service & AI Layer**
3. **Integration & Data Layer**

Each layer consists of 6 sub-modules performing core functions.

Layer	Sub-function Module	Main Role
UI/Controller	User Interface Module	Collect user input, control screens, final approval, guide results
Core Service & AI	Voice & NLP Module	STT, intent classification, extract date/time/menu/price

Core Service & AI	Store Management Module	Manage standard store info, update commands, and business rules
Core Service & AI	Review Analysis & Rec Module	Analyze review sentiment/keywords, recommend replies/copy
Integration & Data	External Platform Integration	Convert platform data, reflect info, manage results
Integration & Data	Data Storage Module	Store user/store/analysis results/history/task status

(Detailed breakdowns of Layer 1, 2, and 3 were translated as outlined in the original document, emphasizing the flow from UI collection -> AI Processing (creating OperationData) -> Backend execution (Redis/Platform Adapters).

5.2 Development Environment

Developed with a separated frontend/backend structure suitable for mobile and AI/API integration.

- **Mobile Frontend:** React Native, Expo, TypeScript, Expo Router, TanStack Query.
- **Backend Framework:** FastAPI (Python), Pydantic.
- **AI/ML:** NAVER CLOVA Speech, KoNLPy, KoBERT/KcBERT, CLOVA OCR API, LLM API.
- **Database & Infrastructure:** PostgreSQL, SQLAlchemy, Redis, Celery (Task Queue), Docker, Nginx, AWS ECS Fargate (Under review).

MVP Development Priorities:

1. **Phase 1 (Core):** STT integration, intent/slot extraction, OperationData generation, DB storage, Google Business Profile API sync, Naver/Kakao guides.
2. **Phase 2 (Quality):** OCR correction, KoBERT sentiment classification, Redis/Celery async retries, LLM copy generation.
3. **Phase 3 (Expansion):** RAG Chatbot, competitor analysis, cloud deployment.

5.3 System Core Function Definition

(Detailed Use Cases UC1 - UC6 translated maintaining business logic flows, alternate scenarios, and platform specific handling).

- **UC1:** Voice Modification of Store Operating Info
- **UC2:** Information Reflection Status Monitoring

- **UC3:** Review Analysis & Marketing Copy Recommendation
- **UC4:** Automatic Store Registration via Business License
- **UC5:** Map Platform Auth Integration
- **UC6:** Help Chatbot Inquiry

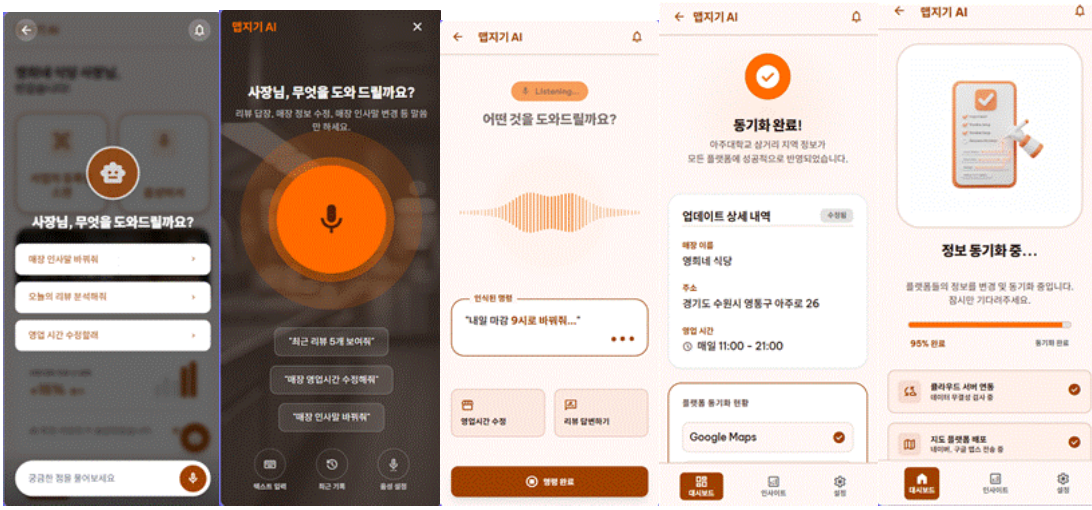
5.4 Basic UI Design & UML Diagrams

5.4.1 Basic UI Design

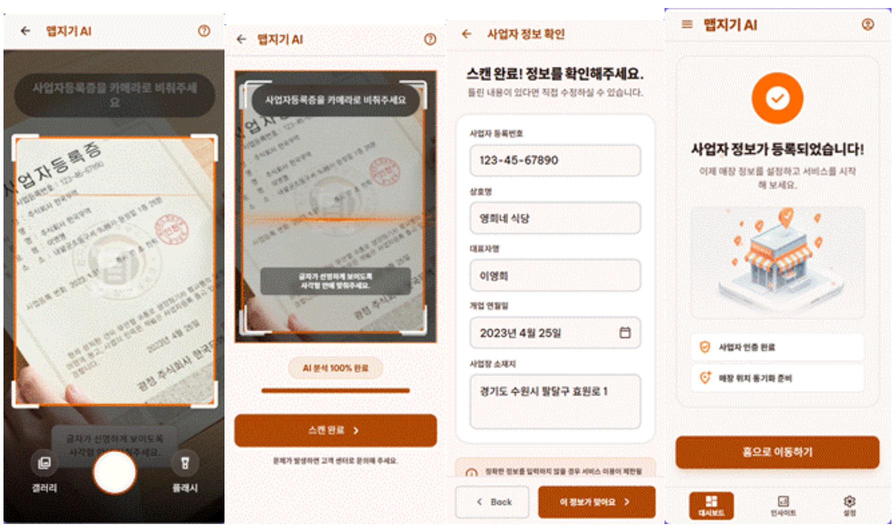
Designed with 'Zero-UI' to minimize complex text input and screen depth for the 5060 target audience.

- **Main Home Screen:** Large buttons for 'Business License Scan' and 'Voice Assistant'.

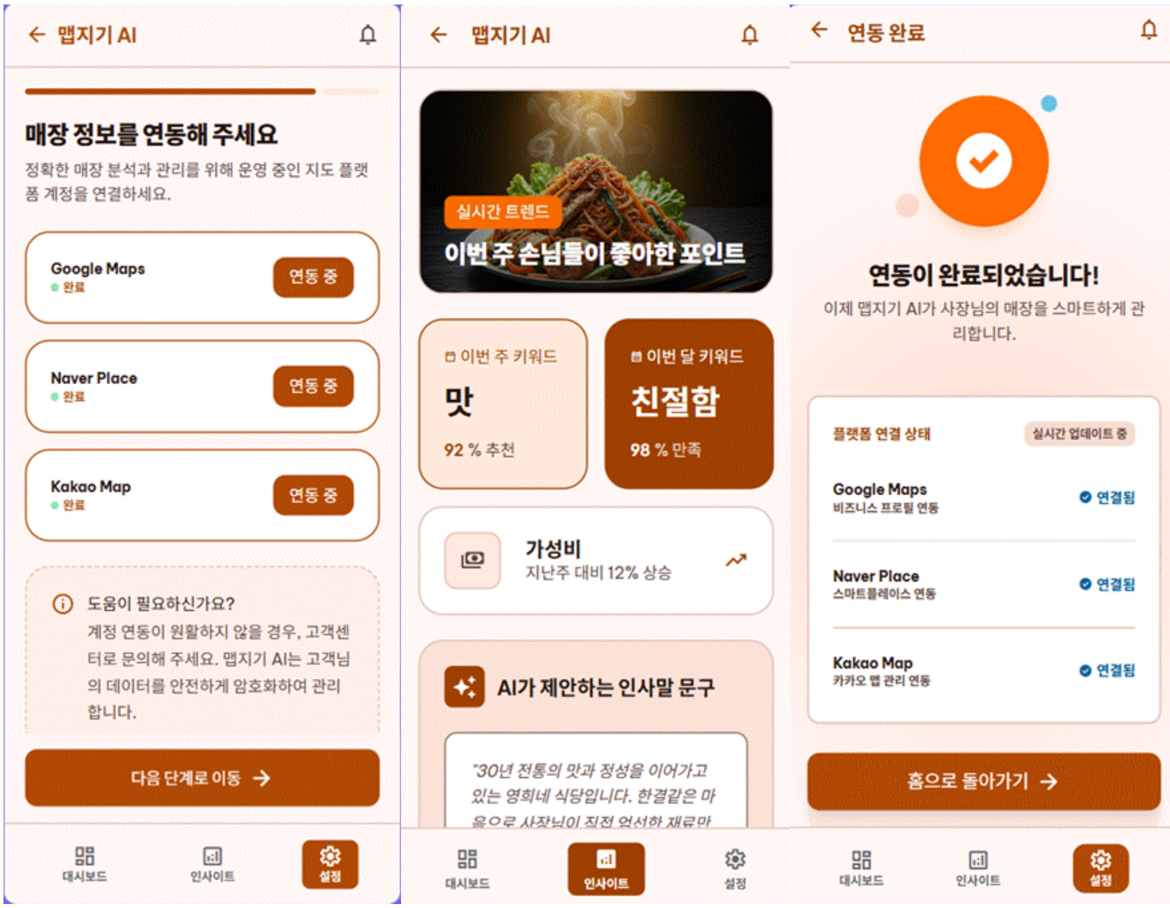
- **Assistant AI & Voice Recognition:** Visual wave animations during STT.



- **Business License Scan:** Camera-guided OCR with confirmation forms.



- **Insight Provision:** Visualized keywords, AI greetings, and 3-platform sync status.



6. References

- Small Business 365: Ministry of SMEs and Startups, 'Big Data-based Small Business Management Support Platform Enhancement Project' (2024~2025).
- Market Analysis: Yext Investor Presentation 2024.
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